

1/13/2026: Extract from RW0 introductory lecture to
Astronomy 8500, Spring 2026

II. ASTR 8500 History since 2016

Have offered this course 5 times since 2016. This is the
6th time.

Basic message of 5 earlier offerings; [Spoiler alert!]

You've CHOSEN THE RIGHT PROFESSION. THERE'S GREAT PROMISE,
SCIENTIFICALLY AND CAREER-WISE.

Science of astronomy doing spectacularly well: all kinds of new
capabilities -- SUPER TECHNOLOGY.

Observational side: giant observatories on the ground, powerful
observatories in space, (studying atmospheres of planets around
other stars), a radio interferometer the size of the Earth (can
resolve SMBHs), neutrino observatories, gravitational wave
observatories....

[Many astronomers expected in the 1980s that we would
eventually detect exoplanets around other stars. But very
few predicted -- or even speculated -- that we would be able
to analyze their _atmospheres_ as soon as we did.]

Computational side: unprecedented analytical power in computers --
especially in simulating astrophysical processes in realistic
detail not possible before -- cosmology/galaxy evolution;
formation of planets; etc.

On the verge of answering many of the questions that people in my
generation thought wouldn't be possible for another century or
more: cosmology, formation and evolution of galaxies, exoplanets,
formation and evolution of planetary systems, extraterrestrial
life....

The term "GOLDEN AGE" does apply.

AND: on top of that, the job market was much better than many grad
students believed...

TWO THIRDS of PhDs are working directly in astronomy doing
research and/or teaching. The other third work in related

high-tech fields -- e.g. supporting the space program if not strictly astronomical missions, high-end computing (AI coming), scientific imaging, instrumentation, etc.

BUT! You don't get this for free! Must put in the work...

Grad students need a number of SKILLS they don't have to start with. Must DELIBERATELY WORK TO CULTIVATE THOSE: work habits, writing skills, persuasiveness in writing papers & funding proposals, knowledge of the literature, networking with other scientists, time management, getting along with other people, and so on.

Have to PAY ATTENTION to career issues in ways that people didn't 40 years ago.

Take a CONSCIOUSLY STRATEGIC approach to career issues.

**But the BASIC MESSAGE WAS POSITIVE: IF YOU PUT IN THE WORK, THE CHANCES FOR A GREAT CAREER IN ASTRONOMY ARE EXCELLENT.

=====

III. So, that was the message I initially thought we'd be sending once again in 2026.

BUT SOMETHING HAPPENED last year which caused some of us to WONDER WHETHER THERE WOULD EVEN BE A PROFESSION OF ASTRONOMY IN THE US IN THE FUTURE!!

WHAT WAS THAT SOMETHING??

Yes: the date was January 20, 2025 and the something was the Inauguration of Donald Trump.

The Trump administration came in, and they immediately started scrapping people and programs left and right. At first, I tended to give them the benefit of the doubt...I thought this was just the MINDLESS THRASHING of people trying to cut budgets quickly without any planning or knowing what they were doing...In other words, they had the excuse of "ignorance."

...But I don't think that anymore. It's now pretty clear that there was a DELIBERATE PLAN TO UNDERMINE SCIENCE AND HIGHER

EDUCATION. It was a deliberate SCORCHED EARTH POLICY.

[Instead of a prudent investment policy, based on evaluating strengths and weaknesses, it adopted a scorched earth policy.]

And it wasn't just fields that seemed to have no practical applications -- like "useless" astronomy (sad to say -- we have to admit this).

No. They cut \$500M from messenger RNA research -- the incredibly promising technology for treating and preventing all kinds of disease. They froze billions of dollars of other National Institutes of Health grant programs. And they ravaged other support for public health, critical renewable technologies, education from preschool to graduate school, and more.

Two important aspects to this:

FIRST. THERE IS NO PEACETIME PRECEDENT FOR WHOLESALE CUTTING OFF FUNDS FOR ALREADY-APPROVED RESEARCH ALLOCATIONS.

It's almost CERTAINLY ILLEGAL. There are many rules regarding hiring and firing, grants and contracts. You can cancel a grant for CAUSE -- but there's a long procedure of negotiation involved. The administration SIMPLY IGNORED ALL THAT and just cut the funds. Can be remedied, but the courts, as always, are slow to respond. A lot of damage is done in the interim.

SECOND, and more important: this was a COMPLETE DISCONNECT FROM THE TRADITION OF GOOD FEDERAL SUPPORT for basic scientific research that had been the norm for ALMOST 80 YEARS -- since 1948 or so. It was at that time the government realized how much basic science had contributed to winning World War II and how essential it had become to our national priorities. Realized it had to support science across the board as a MATTER OF NATIONAL SECURITY AND WELL-BEING.

So the government entered a SOCIAL CONTRACT with the private sector: the government would fund BASIC research while the private sector would convert that research into practical applications -- i.e. useful TECHNOLOGY.

And that eventually established a basic level of federal support for science (and higher education). There was always scatter or jitter in that as administrations and priorities changed, but the basic principle never changed.

This policy worked BRILLIANTLY and made the US the WORLD LEADER in both science and technology.

[In terms of discoveries, technology development, patents, etc.]

THE NASA BUDGET: is a good representation of federal PRIORITY for basic research in science. Show slide.

NASA came on the scene with a huge surge of funding in early 60's. What caused that? Does anybody know? [Hint: An unprecedented POLITICAL AND TECHNOLOGICAL HUMILIATION for the USA.]

SPUTNIK! October 4, 1957 -- a date everybody should know. The beginning of the space age: the first orbiting satellite. And it was the USSR's.

The shock CONFIRMED the policy already in place to support science -- but demonstrated it was inadequate.

After the Apollo program, NASA support for space science was good, but almost FLAT IN CONSTANT DOLLARS. I.e. the ability to buy goods and services has remained fairly constant and even increased to a new plateau after the Challenger accident and in the era of the "Great Observatories" (ca. 1988 plus: HST, Compton, Chandra, Spitzer).

But with the improvement in technology over the last 50 years, that means it would support MUCH MORE EFFECTIVE SCIENCE as time went on -- even if it didn't increase in proportion to the number of space scientists.

[PPT]

The Trump admin BROKE the social contract and PROPOSED HUGE, UNPRECEDENTED CUTS in basic research -- the most relevant to us being in the NSF and NASA.

47% cuts in NASA science budget
55 operating and planned missions terminated

57% cut to NSF

REF: the whole sorry story is documented in the "news clips" file linked to A8500 home page if you care to read the gory details.

The scientific community was STUNNED at first and wasn't sure how to respond...but eventually rallied. Eventually generated tremendous resistance, blowback...and CONGRESS THANKFULLY, did NOT GO ALONG WITH THE PROPOSED CUTS TO FUTURE BUDGETS.

The funding situation only got clarified in the last 10 days.
[SHOW: Congressionally approved budgets.]

Things look better now, but the situation is FRAGILE -- because underlying ANIMOSITY toward science/HED is still there and the administration is not only ERRATIC but malicious in some ways.

The department and the College of Arts and Sciences are (I believe) committed to maintaining support for admitted students.

**An important point -- just between us...I think you shouldn't discuss this with older grads: As first and second year grad students, you're in a much better position than older students who are almost getting their PhDs. Whatever happens -- and it won't be clear for some time which way things will go -- you will have time to ADJUST AND TO PLAN to the changes. That's not true for older people.

From the linked article on "There Has Never Been a Better Time to Start a Ph.D.":

"This is a dire moment to enter the work force.

"It is a perfect moment to spend four to seven years acquiring rare and valuable skills before entering a work force which will not be, in five years, what it is now."

...perhaps an exaggeration but the sense of the situation is right.

Early last year I had asked myself, "God! What am I going to say to the students? The world is crashing down! Doesn't this upend everything?" -- but thinking a little more carefully, the answer is NO. I WOULD GIVE STUDENTS BASICALLY THE SAME ADVICE AS BEFORE, just a little more emphatically.

The advice is to focus more strongly on your education and career planning.

You don't have to think about career issues every day...but half an hour every week or two might be good.

Knowledge? Skills? Milestones?

A lot of the elements you'll need to take into account are covered in my "Tips for Success in Observational Astronomy" (posted on the webpage).